

[This question paper contains 6 printed pages.]

Sr. No. of Question Paper : 5701

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Your Roll No.....

Unique Paper Code : 234101/251305

Name of the Paper : Programming Fundamentals (CSHT-101)/Fundamentals of Programming Languages – CS – 1

Name of the Course : ~~B.Sc. (H) Computer Science~~ / Electronic Science

Semester : ~~I~~ / III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. There are two parts of the question paper.
3. **Part I** : All its questions are compulsory.
4. **Part II** : Attempt any four out of six.

Part I

Question No. 1 is compulsory.

1. (a) What is the type of following constants ?

Unsigned long
Long double

(i) 86392LU

(ii) "2"

String

($\frac{1}{2} \times 4$)

(iii) 7.7623465L

(iv) -6.0f

float

(2)

- (b) Find errors in the following C++ code segments and write the correct code. (6)

(2)

(i) switch (n) {

case 1.1: cout << n * 1.1; break;

case 5.6: cout << n * 5.6; break;

default: cout << n;

}

P.T.O.

```
(ii) int func(int *a, int *b) {
        return ( *a++ + *b++);
```

(2)

```
    }
    void main() {
        int x = 2, y = 7;
        cout << func(2, 7);
    }
    2x, 2y
```

```
(iii) class Test {
        int me;
        char you;
        Test(int a, char b) {
```

(2)

```
            me = a;
            you = b;
        }
    public: ~Test() {}
            void display() {
                cout << me << endl << you << endl;
            }
    };
    (An arrow points from the ~Test() destructor to the display() method.)
```

(c) Write output of the following code segments :

```
(i) int func1(int a, int b) {
        return (a++ + b--);
    }
```

(3)

```
int func2(int& a, int& b) {
    return (++a + --b);
}
```

48

49

```
void main() {
```

```
    int x = int('0'), y = int('5');
```

```
    cout << x << "\t" << y << "\t" << x + y << "\n";
```

(1)

```
    cout << x << "\t" << y << "\t" << func1(x, y) << "\n";
```

(1)

```
    cout << x << "\t" << y << "\t" << func2(x, y) << "\n";
```

(1)

```
}
```

ASCII code for '0' is 48

```

(ii) void main() {
    int a = 20;
    int b = 50, c;
    c = (a != 10) || (b = 30);
    cout << a << '\t' << b << '\t' << c;
}

```

11 ~~20~~ 50 (3)

20 ~~50~~ 1
50

(d) Write output of the following program.

(3)

```

#include <iostream>
using namespace std;
int main(){
    int x = 987;
    cout.width(10);
    cout << x << "\n";
    cout.setf(ios::right, ios::adjustfield);
    cout << x << "\n";
    x = -x;
    cout.fill('0');
    cout.width(10);
    cout.unsetf(ios::adjustfield);
    cout.setf(ios::internal);
    cout << x << "\n";
    x = -x;
    cout.setf(ios::hex, ios::basefield);
    cout << x << "\n";
    return 0;
} // end of main()

```

x x x x x x x 987

(e) What is function overriding in C++ ? Explain with example.

(3)

- (f) List and explain with examples, any two functions that determine the status of a file stream. (3)

- (g) Write a method,

Matrix transpose(); (2)

of a class Matrix, which returns the transpose of the calling Matrix object. Assume the Matrix class consisting of 2-dimensional array and a constructor that accepts size of the matrix (row and columns). (1) (4)

- (h) Write output of the following: (4)

(i) #include <iostream.h>

#include <string.h>

int main() {

char str[] = "Computer";

int len = strlen(str); 9

for (int i = 1; i <= len; i++)

cout << char(str[len-i]+2);

return 0;

}

tgvwroqE

(ii) int x = 0;

while (1) {

if (x%2) {

cout << x++ << '\t';

continue;

}

if (++x >= 10) break;

}

1
3
5
7
9

- (i) Write a code to handle divide by zero exception using try-catch statements. (4)

2. (i) Write a program to copy a file, inserting a blank line after each line. (5)
- (ii) Write a function to sort an array using any algorithm. (5)
3. (i) What are the differences between the following two statements :
char *str1 = new char[10];
char str2[10]; (4)
- (ii) Create a class Employee having a protected, integer id, a default constructor, and a constructor with an argument (for id). Create a derived class SalaryEmp to make the inheritance protected. The SalaryEmp class has a private data member – salary. Write the appropriate inherited constructors to handle all possibilities. (6)
4. (i) Find error(s) in the following code segment.
void main() {
enum color { White, Blue, Pink, Red, Green};
color a, b, c, d;
a = Green;
b = blue;
c = orange;
d = Pink;
}
(2)
- (ii) Write a function to overload += operator for a fraction class using 'this' pointer. Write the fraction class definition consisting of data members, numerator, denominator, and member function print() to display the fraction. (2) (1) (5)
- (iii) Write a copy constructor for the fraction class. Use the copy constructor in main() function. (3)
5. (i) Write a recursive function to calculate GCD (Greatest Common Divisor) of two numbers. (4)

- (ii) Write a program to count the number of digits and alphabets in a given file. Check for the file open and close errors also, in the program. (6)

6. (i) What would be printed from the following block? (5)

```
void main() {
    char* a[5] = {"One", "Two", "Three", "Four", "Five"};
    cout << a[1] << "\n";
    cout << *(a+3) << "\n";
    cout << *(a[4]+2) << "\n";
    cout << (*(a + 2) + 2) << "\n";
    cout << **a << "\n";
}
```

Two
four
v
r
0

- (ii) What is an Abstract class? Explain the concept with the help of an example. (3)
- (iii) What is the difference between a virtual function and a pure virtual function? (2)

7. (i) Write a program to keep a count of number of objects of a class, created and destroyed using static data members. Also, write a static member function to print the count. Call this function from main() to show the usage. (5)

- (ii) Write a friend function to overload * operator that multiplies two complex numbers and returns the result as a complex number. (5)